# Aishi Tiwari

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LeetCode Portfolio Github

## EDUCATION

**Guru Gobind Singh Indraprastha University (MAIT)** **2024 – 2028**
*B.Tech - Computer Science and Technology -* ***CGPA: 9.10/10*** *Delhi*

**Sardar Patel Vidyalaya (Class XII)** **Completed 2024**
*CBSE, Science Stream (PCM) -* ***SCORE: 90.0*** *Delhi, India*

## EXPERIENCE

**Indian Institute of Technology Delhi with Larsen and Toubro** **May 2026 – ongoing**
*Machine Learning Research Intern* *On-Site*

- Developed a **training-free, foundation-model-guided expert routing framework** for adaptive inference across heterogeneous data distributions, eliminating the need for learned gating networks or additional fine-tuning.
- Designed a semantic similarity-based routing algorithm using DINOv3 embeddings, prototype representations, and cosine-similarity retrieval to dynamically select specialised models.
- Achieved state-of-the-art **55.55 mAP50** on BDD100K through adaptive expert routing, demonstrating training-free model selection across heterogeneous data distributions.
- Submitted research work to **AAAI 2027 & ICLR 2027** main track

**NITI Aayog, Government of India** **July 2025 – September 2025**
*Science and Technology Intern* *On-Site*

- Developed an **end-to-end geospatial ML pipeline** using high-resolution satellite imagery to detect and map healthcare infrastructure across **112 Aspirational Districts** in India.
- Trained a **YOLOv8 object detection model** on a self-annotated dataset of **1,800+ satellite images**, using Roboflow for annotation and dataset management
- Generated infrastructure-profiling outputs identifying healthcare facilities and potential infrastructure gaps to support **data-driven governance and resource planning**.

**TechXtract** **August 2026 – ongoing**
*Technology Head* *On-Site*

- Led the development of a centralised data management platform serving 100+ students and faculty, implementing real-time synchronisation, secure data workflows, and role-based access control using Firebase.

## PROJECTS

### Council- a Multi Agent Orchestration System **September 2026-Ongoing**

- Currently developing **a multi-agent framework** enabling heterogeneous LLMs to collaboratively construct, critique, and iteratively refine solutions for mathematical reasoning and English (writing) tasks.
- Implemented role-specialized **Builder, Critic, and Judge agents** with shared state and iterative cross-agent feedback.
- Integrated **multiple LLM APIs** and developed a unified layer for **cross-model communication** and response aggregation.

### Helium – Compiler for a Custom Programming Language **July 2026 | GitHub Link**

- Developed **Helium**, a compiler in C++ that generates x86-64 Linux executables using NASM and LD.
- Designed a modular compiler architecture with a hand-written recursive-descent parser, cleanly separating lexical analysis, parsing, and code generation.
- Implemented **10+ core language features**, including conditionals, functions, return statements, and I/O, with correct assembly-level control flow without relying on compiler frameworks or backend libraries.

## TECHNICAL SKILLS

**Machine Learning and AI:** PyTorch, Scikit-learn, Keras, Hugging Face, YOLO, Roboflow, NumPy, Matplotlib, CUDA
**Programming Languages:** Python, C++, C, Java, SQL, JavaScript, TypeScript
**NLP/LLM:** RAG, LangChain, Hugging Face Transformers, spaCy, NLTK, vector DBs (FAISS/Chroma)
**Development & Tools:** Linux, Git, GitHub, NASM, LD, Firebase, Flask, FastAPI, React, Node.js, HTML, CSS
**Data Structures and Problem Solving:** Solved 350+ problems on LeetCode along with regular Contest participation.

## ACHIEVEMENTS & CERTIFICATIONS

- 5X Hackathon finalist +2X winner at inter and intra-college levels+ **Finalist Smart India Hackathon 2025**
- **Stanford University**– **Probability for AI**: Selected for the competitive course focused on the mathematical foundations of AI.
- **SheFi – 2025 Cohort Scholar**: Selected for a competitive Web3 and AI program. **Recipient of an $800 scholarship.**
- **Open-source**: day/night YOLOv11 detectors on BDD100k dataset, published on Hugging Face (mAP50 56.6 (day) / 50.0 (night))
- **Appointed Technology Head of TechXtract- The Official Society of the Computer Science Department.**
- Among the **top 1,500 programmers in Google** Big Code, a competitive nationwide coding challenge.
- **Certifications**: Andrew NG's Machine Learning Specialisation (Coursera, DeepLearning.AI), Machine Learning Learning Plan (AWS) Advanced MCP+ Introduction to MCP (Anthropic).